Cover letter

Dear Editors,

We are resubmitting *GENESIS 2.5: optimisation and opt-in OpenCL/CUDA acceleration for the GENESIS/PGENESIS compartmental neural simulator*, previously handled as SOFTX-D-26-00952.

The earlier decision turned on the absence of a head-to-head comparison with other simulators and on questions about the benchmarking methodology. Rather than argue the point, we went and did the measurements. The paper is substantially different as a result.

Head-to-head, on two models, on CPU and GPU. GENESIS 2.5 is now compared with NEURON, CoreNEURON and Arbor on the published Vogels-Abbott COBAHH network, and with NEURON and Arbor on a 160,000-compartment multi-compartment population. The competitors' GPU backends are included: CoreNEURON through its OpenACC path and Arbor through CUDA. We say why those three and not NEST or Brian 2 — they simulate point-neuron networks rather than dendritic morphologies, so they would not be running the same computation.

Equivalence verified rather than assumed. Independent implementations of the same published benchmark differ. We compared recorded membrane potentials across the three simulators before comparing their wall times, and report where they disagree and why. Setting that up exposed three errors in our own harness, which we corrected and disclose.

One timing definition throughout. Every ratio in the paper wall-clocks both arms around the whole process, construction and transfers included. We state this explicitly because timing the two arms by different clocks inflates speedups roughly tenfold, and we report the step-phase figures separately rather than quoting them as what a user gets.

Results that do not favour us are in the paper. CoreNEURON on an A100 runs the spiking benchmark 1.23x faster than our best arm. The accelerator now runs a synaptically coupled spiking network — new in this release — but only at parity with its own CPU arm, and we explain why: a zero-delay network cannot batch dispatches. Neither our simulator nor Arbor is faster in general; their lines cross, and where they cross depends on the card, which we measured on two.

A reproduction pack. `sh reproduce/run_all.sh` rebuilds the software, re-measures the claims on the reader's own hardware, regenerates the figures from those measurements, and prints each published value beside theirs with a verdict. It refuses to time a binary built for a different GPU than the one in the machine — a failure that cost us a day and would otherwise cost a reviewer one.

On the Code Ocean capsule. The submission form asks for one. We have not made a capsule, and we would rather explain why than leave the field empty. What this paper claims is GPU acceleration, and a capsule without an NVIDIA card would exercise only the CPU paths — demonstrating nothing that is in dispute, and possibly suggesting the results cannot be reproduced. In its place we offer the repository, an archived release whose DOI names the exact version every number was measured on, and the reproduction pack described above, which runs on the reviewer's own hardware rather than on a machine we chose. If the editors would still like a capsule, we can prepare one for the two claims that need no GPU: model construction reduced from $O(n^2)$ to linear, and the 1.41x gained on the CPU by building a spiking network as one solver per layer.

The software is archived at https://doi.org/10.5281/zenodo.22032886, which is the exact version the measurements were taken on.

We hope the revised manuscript now meets the standard the reviewers asked for.

Yours sincerely,

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